Phase 9 CVRP Solver Evolution Report

Overview

- Condition count: 4.
- Best held-out condition: phase9_baseline_clarke_wright_savings.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Mean Novelty	Mean Complexity
phase9_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	42.75206	0.0	0.0
phase9_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	77.4275	0.0	0.0
phase9_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1465.37042	0.0	0.0
phase9_solver_evolution	solver_evolution	1.0	0.273435	0.273435	141.1843	0.0	0.0

Condition Notes

phase9_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 42.75206.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.064783.

- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 77.4275.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1465.37042.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:
- X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_solver_evolution

- Execution mode: solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 141.1843.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]

- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

Judge Note

Phase-9 CVRP Solver-Evolution Suite Summary

Solver Condition	Feasibility Rate	Heldout Feasible Gap	Heldout Runtime (ms)	Robustness Dispersion	Notes on Performance
Baseline Nearest Neighbor Constructive	1.0	0.2734	42.75	0.00278	Fastest runtime, moderate gap
Baseline Clarke-Wright Savings	1.0	**0.0738 (best)**	77.43	0.00779	Best solution quality (lowest gap), moderate runtime
Baseline Regret Insertion Local Search	1.0	0.4664 (worst)	1465.37	0.0869	Longest runtime, worst gap, least robust
Phase-9 Solver Evolution	1.0	0.2734	141.18	0.00278	Same gap as nearest neighbor, slower runtime, very robust

Conservative Conclusions

- **Feasibility:** All solvers maintained perfect feasibility (1.0).
- **Objective Gap:** The evolved solver's gap (0.2734) matches the nearest neighbor baseline but is notably worse than Clarke-Wright (0.0738).
- **Runtime:** Evolved solver runs slower than nearest neighbor (141 ms vs. 43 ms) but much faster than regret insertion (1465 ms).
- **Robustness:** The solver evolution approach exhibits very low robustness dispersion (0.00278), equal to nearest neighbor and better than other baselines.
- **Overall:** The evolved solver does *not* clearly outperform the best fixed baseline (Clarke-Wright Savings) in terms of solution quality or runtime. It roughly matches the nearest neighbor baseline's gap, at a cost of slower execution.

Summary: The evolved solver is feasible and robust but does not clearly beat the best fixed baselines on held-out structure families in objective gap or runtime.